

Can Language Models Forecast Their Own Belief Updates?

A Two-Context Test of Metacognitive Forecast Revision

Invariant Forecasting Project

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Abstract

Probabilistic forecasts are useful only insofar as they can be revised under new evidence. Recent work studies whether language models can forecast future events, whether they update beliefs in a Bayes-like manner, and whether they can predict aspects of their own future behavior. We connect these threads with a direct metacognitive test: can a language model predict how its own forecast will change after seeing evidence? For each forecasting question X and neutral evidence packet Y , we first elicit the model’s current probability $\pi = P(X)$. In the same context, before allowing an ordinary update, we ask the model to predict what its own posterior would be after receiving Y . In a fresh context, we then actually provide Y and elicit the realized posterior. We compare predicted and realized updates in log-odds space. On the full 343-question ForecastBench-derived validation split using GPT-OSS-120B, the model’s predicted and realized log-odds updates have Spearman correlation 0.908 and Pearson correlation 0.894, reducing mean absolute log-odds error by 65.4% relative to a no-update baseline. The result is stable under robustness checks: restricting to $|\Delta_{\text{act}}| < 5$ gives Spearman 0.907 and Pearson 0.886. Error analysis shows that failures are concentrated in a small number of near-boundary probability cases, while ordinary-sized updates are strongly predictable. These results suggest that language models exhibit measurable self-forecasting of their own evidence-conditioned forecast revisions, without implying Bayes optimality or privileged access to internal beliefs.

1 Introduction

Forecasting systems should not merely issue probabilities; they should also revise them coherently when new information arrives. Large language models (LLMs) are increasingly evaluated as forecasters on real-world questions [1, 2]. Separately, a growing literature asks whether LLMs can evaluate their own knowledge or anticipate their own behavior [3, 4, 5], and another asks whether LLMs can perform probabilistic or Bayesian-style belief updating [7]. What is missing is the direct bridge: can a model predict its own future belief update before the evidence is actually shown in an ordinary forecasting context?

We propose a simple two-context protocol. Given a binary question X , first elicit the model’s current forecast π . Then, in the same context, present a counterfactual description of evidence Y and ask the model to predict the posterior it would give after incorporating Y . Finally, in a fresh context, provide the same evidence Y and elicit the model’s actual posterior. The central object is not the posterior itself, but the update:

$$\Delta_{\text{pred}} = \text{logit}(\pi'_{\text{pred}}) - \text{logit}(\pi), \quad (1)$$

$$\Delta_{\text{act}} = \text{logit}(\pi'_{\text{act}}) - \text{logit}(\pi). \quad (2)$$

If models have metacognitive access to their own forecast revision behavior, then Δ_{pred} should track Δ_{act} across many (X, Y) pairs.

This paper reports an initial, deliberately narrow result: neutral evidence packets only, one model, and the full 343-question validation split. The result is nonetheless sharp. Predicted and actual updates are strongly rank-correlated, substantially closer than a no-update baseline, and robust after excluding exact boundary probabilities. The remaining failures are also interpretable: they are mostly near-boundary probability cases or directional disagreements on individual questions, rather than a broad collapse of self-prediction.

2 Protocol

Each example consists of a binary forecasting question X , a prior phase, and a neutral evidence packet Y dated before the forecast date. The prior phase includes the question, source background, resolution criteria, forecast date, resolution date, and a reference-class base rate. The base rate is shown as context, but in the exact protocol it is not used as π ; π is elicited from the model.

Step 1: elicit the current forecast. The model receives only the prior phase and is asked to output a probability on the first line in the form `Probability: <number between 0 and 1>`. This probability is π .

Step 2: predict the future update. In the same conversation, the model is told that it is about to receive evidence Y in a separate ordinary forecast. Before making that ordinary forecast, it is asked to predict what its own posterior forecast would become after incorporating Y . This yields π'_{pred} .

Step 3: realize the update in a fresh context. In a new context, the model receives the same prior phase, acknowledges that it will use the prior phase as its starting point, then receives Y as the update phase and outputs a posterior. This yields π'_{act} .

Metric. We compare log-odds updates from the same elicited prior. To avoid infinite values, probabilities are clipped to $[10^{-6}, 1 - 10^{-6}]$ before applying $\text{logit}(p) = \log(p/(1 - p))$. We report Pearson and Spearman correlations between Δ_{pred} and Δ_{act} , mean absolute error (MAE), RMSE, posterior probability gap, and improvement over a no-update baseline ($\Delta_{\text{pred}} = 0$).

3 Data and Model

We evaluate on all 343 validation questions from the repository’s ForecastBench-derived dataset. The evidence packets are the neutral variants of the generated prior/update dataset; stronger weak-yes, strong-yes, weak-no, and strong-no framings are reserved for future experiments. We run GPT-OSS-120B via Tinker sampling at temperature 0, top- $p = 1$, and a 512-token generation cap. The run produced 342 parseable prior forecasts and 330 fully parseable prior/predicted/actual triples, giving pair coverage 96.2%.

4 Results

Table 1 gives the main result. Across all parseable pairs, predicted and actual log-odds shifts have Spearman correlation 0.908 and Pearson correlation 0.894. The model reduces log-odds MAE by 65.4% relative to predicting no update. Unlike the earlier 200-group smoke test, the full validation

Metric	All rows	Excl. exact 0/1	$ \Delta_{\text{act}} < 5$
Parseable pairs / rows	330 / 343	330 / 343	329 / 343
Pearson corr.	0.894	0.894	0.886
Spearman corr.	0.908	0.908	0.907
Log-odds MAE	0.289	0.289	0.290
Log-odds RMSE	0.438	0.438	0.438
Posterior RMSE	0.041	0.041	0.041
No-update MAE	0.836	0.836	0.824
MAE improvement vs no-update	65.4%	65.4%	64.8%

Table 1: Main exact-protocol results. The model substantially predicts its own future update. Boundary filtering and a $|\Delta_{\text{act}}| < 5$ restriction leave the main conclusion essentially unchanged.

$ \Delta_{\text{act}} $ bin	n	Pearson	Spearman	MAE	RMSE	Post. MAE
< 1	229	0.874	0.885	0.205	0.311	0.025
$[1, 3)$	99	0.848	0.701	0.476	0.629	0.017
$[3, 5)$	1	–	–	1.386	1.386	0.00015
≥ 5	1	–	–	0.000	0.000	0.000

Table 2: Performance conditioned on actual update extremity. Most examples fall below $|\Delta_{\text{act}}| < 3$, where linear/rank correlations and errors remain strong. The extreme bins are too small for stable correlation estimates.

run contains no exact 0/1 posteriors among parseable pairs, so the all-row and boundary-filtered metrics coincide. Restricting to $|\Delta_{\text{act}}| < 5$ yields nearly identical results.

Figure 1 shows the same pattern visually. Most points cluster around the diagonal, with no exact 0/1 posterior among parseable pairs in the full validation run.

5 Distribution-Conditional Diagnostics

The full validation run supports a distribution-conditional story: the model predicts ordinary updates especially well, while rare extreme updates are too few to estimate reliably. We therefore condition on $|\Delta_{\text{act}}|$ bins.

Table 2 and Figure 2 are the key diagnostic evidence. There are 328 parseable pairs with $|\Delta_{\text{act}}| < 3$. In these ordinary regimes, Pearson correlations are 0.874 and 0.848 in the first two bins, with log-odds MAE below 0.48. The $[3, 5)$ and ≥ 5 bins contain one example each, so they are useful as case studies but not as stable estimates of correlation.

6 Worst-Case Examples

The worst cases are informative rather than merely embarrassing. Several large errors involve small prior or posterior probabilities, where modest probability changes can still produce large log-odds shifts. Others are genuine directional or magnitude disagreements, such as predicting a strong positive update when the fresh-context posterior moves down. This suggests that future runs should report both log-odds-space metrics and probability-space metrics, and should audit near-boundary behavior carefully.

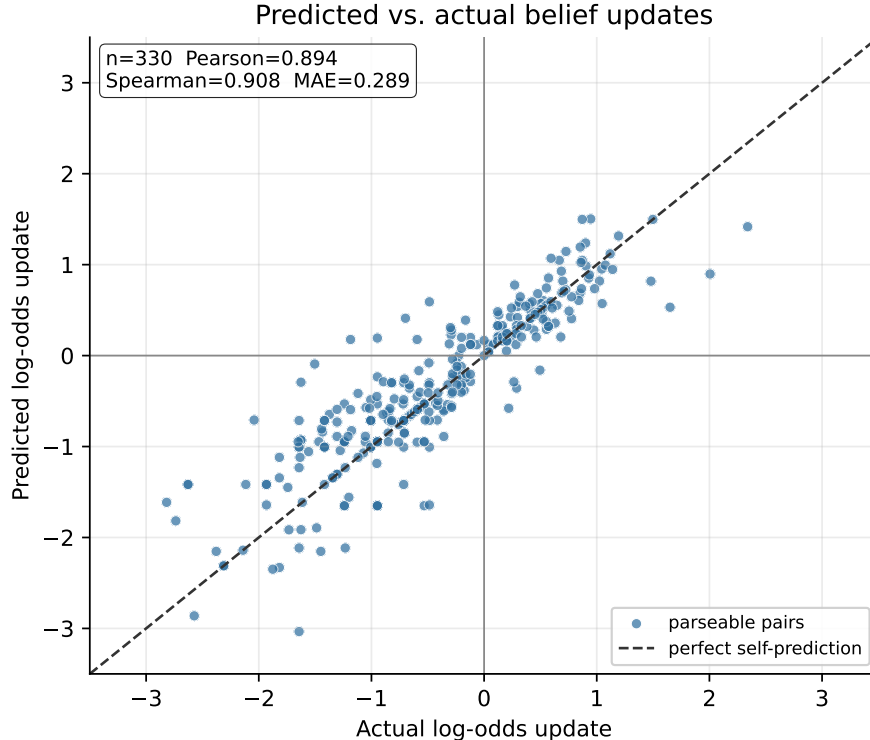


Figure 1: Predicted versus actual log-odds updates in the exact two-context protocol. The dashed line is perfect self-prediction.

7 Related Work

LLM forecasting. Forecasting benchmarks evaluate whether LLMs can produce calibrated probabilities for future events. Halawi et al. show that retrieval-augmented LM systems can approach competitive human forecasting in some settings [1]. ForecastBench provides a dynamic benchmark of future-event forecasts designed to reduce contamination risk [2]. Our work is complementary: we do not primarily evaluate forecast accuracy, but whether a model can predict its own update when evidence is introduced.

Self-knowledge and self-prediction. Kadavath et al. study whether models can evaluate the correctness of their own answers and predict whether they know an answer [3]. More recent work asks whether models can predict their own behavior or learn about themselves by introspection [4, 5]. We adopt the self-prediction lens but apply it to probabilistic forecast revision rather than task success, answer validity, or output behavior.

Bayesian and probabilistic reasoning in LLMs. Bayesian inference gives a normative account of belief revision under evidence. Recent work investigates how LLMs represent priors or learn probabilistic reasoning behavior [6, 7]. Our experiment is not a test of Bayes optimality. Instead, it asks whether the model’s own counterfactual prediction of its posterior update matches the posterior update that actually appears under the same evidence in a fresh context.

Self-prediction remains strong across ordinary update sizes

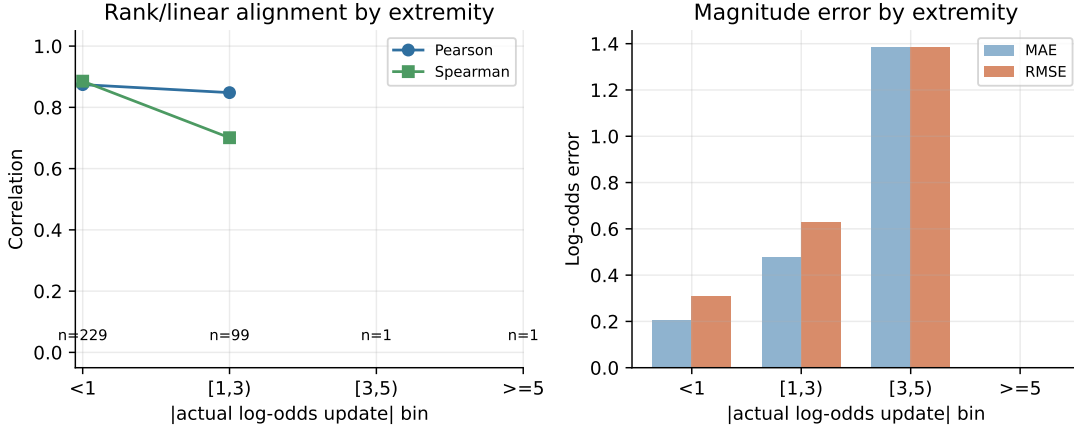


Figure 2: Correlations and error by actual-update extremity. The two-regime picture is visible: ordinary updates are predictable; rare extreme shifts are sparse in this validation split.

Question	π	π'_{pred}	π'_{act}	Δ_{pred}	Δ_{act}
Solana reaches \$100 in March	0.300	0.950	0.680	3.792	1.601
New Coronavirus pandemic in 2025	0.035	0.032	0.008	-0.093	-1.504
Socialist Party wins most seats in 2025 Netherlands election	0.040	0.002	0.008	-3.035	-1.642
Oviedo wins the 2025–26 La Liga	0.003	0.0002	0.00005	-2.711	-4.097
Marburg hemorrhagic fever vaccine by 2025-04-20	0.120	0.140	0.040	0.177	-1.186

Table 3: Top worst-case examples by absolute log-odds shift error. Boundary and near-boundary probabilities can create large log-odds effects even when probability differences are modest.

8 Limitations and Next Steps

This is a workshop-scale draft result with several limitations. First, we evaluate one model and neutral evidence only. The same protocol should be run on smaller open models and on weakly/strongly biased evidence framings. Second, the exact protocol uses generated evidence packets and a particular prompt format; prompt sensitivity remains an open question. Third, log-odds metrics are appropriate for forecast revision but highly sensitive near 0 and 1. We therefore report robust subsets and probability-space gaps, but future work should pre-register clipping rules and boundary handling. Finally, a model’s ability to predict its output under a prompt is not evidence of conscious self-knowledge. The claim is behavioral and operational: predicted update tracks realized update.

9 Conclusion

We introduced a two-context protocol for testing whether language models can forecast their own future belief updates. On 343 neutral ForecastBench-derived examples, GPT-OSS-120B predicts its own evidence-conditioned log-odds updates substantially better than simple baselines. Both rank and linear correlations remain high across the full run, and magnitude errors are concentrated in a small number of case-level disagreements. These findings suggest a concrete and measurable form of metacognitive forecast revision: models can often say, before seeing evidence in an ordinary context,

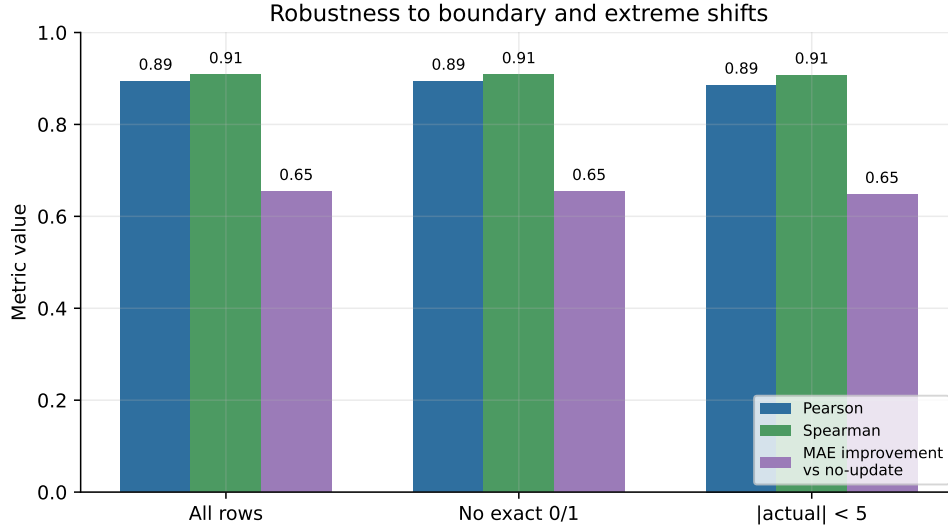


Figure 3: Robustness of the headline metrics. Boundary filtering leaves the full-validation result unchanged, while restricting to $|\Delta_{\text{act}}| < 5$ preserves high Pearson/Spearman correlation and large improvement over no-update.

how that evidence will later move their own forecast.

References

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